# Garthigan Kumarasamy

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Garthigan Garthigan

## Experience/Projects

**Research Assistant** *Colombo,Srilanka*
*Department of Computer Science and Engineering , University of Moratuwa* *July 2025 – Present*

- Conducting deep-learning research under **Dr.T.Uthayasanker**, **Prof. Damminda Alahakoon**, and **Prof. Priyanga Ranasinghe**, in collaboration with La Trobe University,Melbourne and University of Colombo. Developing an image-based system for calorie and nutrition estimation of Sri Lankan mixed meals. Working on food recognition, segmentation, portion estimation, and nutrient prediction using RAG.Using a curated dataset of culturally specific dishes with ingredient and portion annotations given by Prof. Priyanga Ranasinghe . Producing publishable outcomes aimed at computer vision and digital health conferences.

**Machine Learning Engineer** *Colombo,Srilanka*
*Mammilway (Pvt) Ltd.* *Sep 2025 – Present*

- Designing and implementing MLOps workflows on Amazon Web Services (AWS) for AI-driven applications, including LLM-based agents, chatbot services, and recommendation engines, with responsibilities in model deployment, cloud infrastructure, API integration, CI/CD, and service optimization.

**Visiting Instructor** *Colombo,Srilanka*
*Electrical Engineering Department , University of Moratuwa* *Jan 2025 – June 2025*

- As a Visiting Instructor, I instruct and guide undergraduate students in hands-on robotics experiments, focusing on key areas such as sensor integration, motor control, and autonomous navigation. I also assist in designing and enhancing lab exercises to align with current trends in automation and AI-driven technologies, providing students with practical experience that bridges theoretical concepts with real-world applications. My work in this role is carried out under the supervision of **Dr. Manuja Gunawardhana**.

**Research Project** *Colombo,Srilanka*
*Enhanced Multimodel Human Robot Interaction in Service Robots Using Gen AI* *Aug 2024 - Present*
*University of Moratuwa , Srilanka*

- Developed a modular framework for service robots that allows real-time, context-sensitive, and emotionally intelligent interaction using tools such as OpenAI Whisper, YOLOv8, and LLaVA-NeXT-Video. Integrated speech, vision, and emotion recognition, and multimodal fusion implemented for adaptive responses. Multimodal fusion strategies implemented for adaptive and natural human–robot communication. Optimized models through quantization and fine-tuning for resource-limited hardware.This work was carried out under the supervision of **Dr. Buddhika Jayasekara** and **Dr.V Logeeshan**.

**Associate Machine Learning Engineer** *Colombo, Srilanka*
*SenzMate AIOT Intelligencies* *May 2024 – Jan 2025*

- As an Associate Machine Learning Engineer, I conducted an in-depth performance analysis using people detection datasets, leading to improved detection accuracy and refined model optimization strategies. The deployment of the YOLO model with TensorRT on the Nvidia Jetson Nano not only enhanced inference efficiency but also ensured robust, real-time operation. My contributions directly addressed the challenges of running high-performance AI models in edge devices with limited computational resources.

**Machine Learning Engineer - Intern** *Colombo,Srilanka*
*SenzMate AIOT Intelligencies* *Nov 2023 – May 2024*

- During my internship, I developed and deployed a real-time people detection system using the YOLO object detection model on the Nvidia Jetson Nano. By implementing TensorRT optimization in Python, I achieved a significant 30% reduction in inference time, enabling the system to reach a processing speed of 50 frames per second. This optimization made the solution highly suitable for real-time applications in resource-constrained embedded environments.

## Education

**Master of Science (Major Component of Research) - Reading** *Sep 2025 – Oct 2026*
*Department of Computer Science and Engineering*
*University of Moratuwa, Sri Lanka*
**Topic :** Image-Based Calorie and Nutrition Estimation System for Sri Lankan Mixed Dishes using Deep Learning

**BSc in Engineering (Hons), Specialization in Electrical Engineering** *May 2021 – June 2025*
*University of Moratuwa, Sri Lanka*
**GPA : 3.68**
**Relevant Coursework:**

- Robotics and Control
- Programming Fundamentals
- Embedded Systems and Applications
- Embedded Systems Design and Programming
- Industrial Automation
- Pattern Recognition
- Computer Systems
- Digital Signal Processing
- Modular Software Development
- Applied Statistics
- Neural Networks and Fuzzy Logic
- Intelligent Systems and Machine Learning

**Jaffna Hindu College, Jaffna** *Jan 2011 – Aug 2019*
*G.C.E Advanced Level — 3As in Combined Mathematics, Physics, and Chemistry*
*Island Rank: 198, District Rank: 19 , Z-Score - 2.4765*

## Publications

**Towards Enhanced Multimodal Human-Robot Interaction in Service Robots using Generative AI.** May 2025

World AIIoT Congress 2025 , Seattle, May 2025.
Tharmilan, ***Garthigan***, Henoshan

**WebBioAuth: Privacy-Preserving Multimodal Biometric Authentication in the Browser** Sep 2025

CVPR Workshop 2026 - Computer Vision for Biomechanics Workshop , Denver 2026
***K.Garthigan , T.Uthayasanker***

## Awards

**Champions** — Inter School Chemistry Quiz Competition by Samapth Bank *2018*
**Champions** — District Level Badminton Championship by JDBA *2018*
**Second Place** — Video Clip Contest Organized by IEEE Signal Processing Society *2021*
**Participant** — U19 Provincial Level Cricket Championship *2019*
**Participant** — National Mathematics Olympiad by Ministry of Education *2016,2017*

## Skills

**Languages:** Python(Proficient), C++, Java

**Experience and Interest:** Computer Vision, Image Processing ,Deep Learning, Machine Learning ,Robotics and Automation, Human Robot Interaction ,

## Volunteer Experience & Leadership

**President, St John Ambulance — Jaffna Hindu College** *2011 – 2019*
Led first aid training sessions, organized emergency drills and health awareness programs as President of the St. John Ambulance unit at Jaffna Hindu College. Currently continue to serve as a certified First Aid Instructor.

**President, Media Team - University of Moratuwa** *2024 – 2025*
Led media team for event coverage, content creation, and coordination at University of Moratuwa.

**Captain, U17 Cricket Team — Jaffna Hindu College** *2016 – 2017*
Cricket Team of Jaffna Hindu College is known for its competitive spirit and teamwork.Led the U17 cricket team, developing strategies and motivating players to achieve strong team performance.

**Player, Cricket Team - University of Moratuwa** *2022 – 2023*
The University of Moratuwa Cricket Team is recognized for its strong teamwork and competitive performance in inter-university tournaments. Participated in multiple national-level competitions, achieving notable wins. Initiated collaborative training sessions to improve skills and foster team spirit.

**Player, Badminton Team - University of Moratuwa** *2022 – 2023*
The University of Moratuwa Badminton Team is well-regarded for its consistent performance in both singles and doubles events. Competed in several inter-university championships, winning multiple awards. Organized joint practice sessions to enhance technique and teamwork among player

**Director of Design , IEEE Power and Energy Society** *2023 – 2024*
Served as Director of Design for the IEEE Power and Energy Society, leading the creation of visual materials, event branding, and promotional content. Collaborated with teams to ensure professional and impactful design across workshops, seminars, and publications.